Outline

1. Summary
2. Vector space $\mathbb{R}^n$
3. Limits
4. Continuity
5. Partial derivatives
6. The Chain Rules
7. Approximations

Definition

Let $f(x, y)$ be a function defined in a domain $D \subset \mathbb{R}^2$ and $(a, b) \in D$.
The function f is said to be continuous at (a, b) if

$$\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b).$$

Remark. For a continuity of $f(x, y)$ at (a, b), 3 conditions must hold:

1. f is defined at (a, b), i.e. $(a, b) \in D$,
2. there exists a limit $\lim_{(x,y)\to(a,b)} f(x, y) = L$,
3. $L = f(a, b)$.

We say that f is continuous on D if it is continuous at each $(a, b) \in D$.

• Polynomials & Rational Functions.

Definition

- A polynomial of two variables is a function of the form

$$p(x,y) = c_0 + c_1 x + c_2 y + c_3 xy + c_4 x^2 + c_5 y^2 + \cdots + c_k x^n y^m.$$

For example, $p(x,y) = \pi - 19x^2 + 2y^3 - 25xy^9$.

- A rational function of two variables is a ratio of two polynomials $\dfrac{p(x,y)}{q(x,y)}$, with $q(x,y) \neq 0$.

For example, $f(x,y) = \dfrac{x^2 + xy + y^2}{x^2 - y}$, where $y - x^2 \neq 0$.

From the properties of limits, it follows that for non-negative integers m and n, we have

$$\lim_{(x,y)\to(a,b)} x^m y^n = a^m b^n.$$

Thus $f(x, y) = x^m y^n$ is continuous at every point $(a, b) \in \mathbb{R}^2$.

This gives the following important facts:

- Polynomials are continuous on $\mathbb{R}^2$.
- Rational functions are continuous on their domains of definition.

For example, $p(x, y) = \pi - 19x^2 + 2y^3 - 25xy^9$ is continuous everywhere on $\mathbb{R}^2$, while $f(x, y) = \dfrac{x^2 + xy + y^2}{x^2 - y}$ is continuous at any point $(x, y) \in \mathbb{R}^2$ satisfying $y \neq x^2$.

Example

Is the function

$$g(x,y) = \begin{cases} \dfrac{2x^2 - y^2}{x^2 + y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0), \end{cases}$$

continuous at $(0,0)$?

Solution. Note that $g(x,y)$ is defined at $(0,0)$ and $g(0,0) = 0$, and hence we have to check whether $\lim\limits_{(x,y)\to(0,0)} g(x,y) = 0$.

But we already proved that $\lim\limits_{(x,y)\to(0,0)} g(x,y) = \lim\limits_{(x,y)\to(0,0)} \dfrac{2x^2 - y^2}{x^2 + y^2}$ does not exist, and so g is not continuous at $(0,0)$.

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Definition (Partial derivative of $f(x, y)$ w.r.t. x)

For a function of two variables $f(x, y)$, partial derivative of f w.r.t. x is the function f_x defined as follows:

$$f_x(x, y) = \lim_{t \to x} \frac{f(t, y) - f(x, y)}{t - x},$$

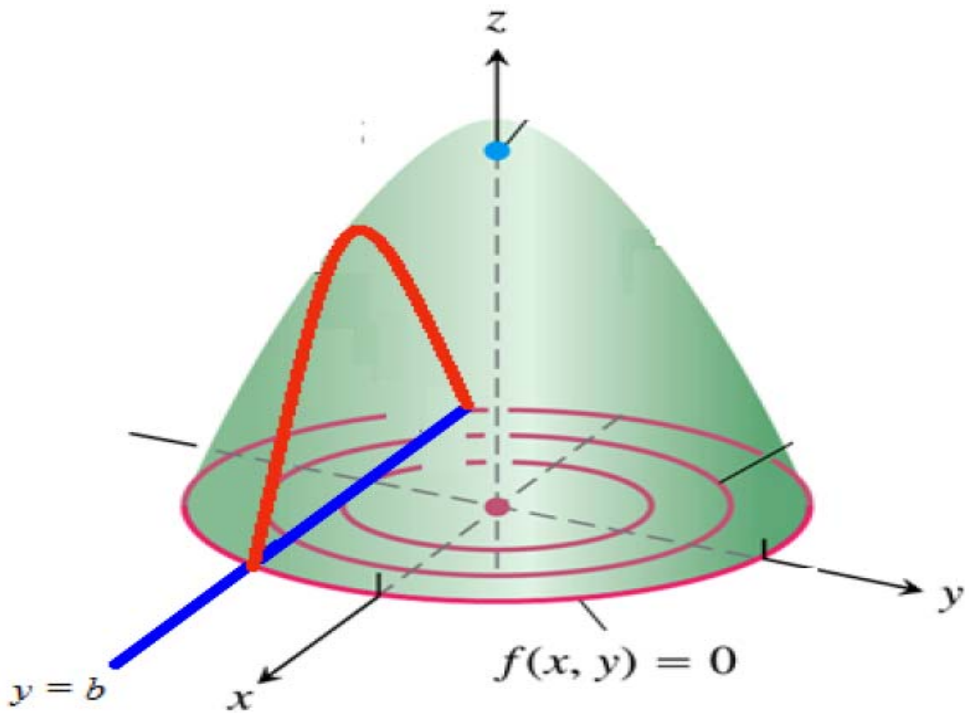
(keeping y as a constant).

Note that the subscript x indicates that the values of t tends to x.

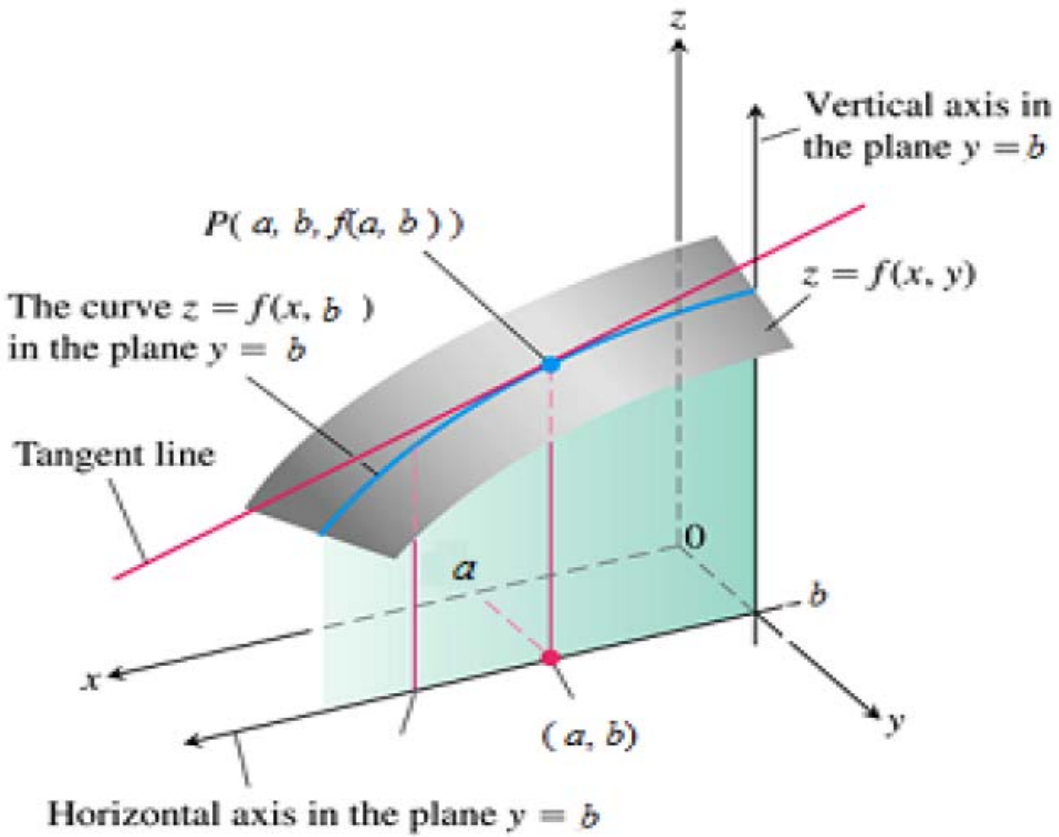
Another notation: $\dfrac{\partial f}{\partial x}$.

The symbol $f_x(a, b)$ is the partial derivative $f_x(x, y)$ at a point (a, b) (i.e., $x = a$, keeping $y = b$ unchanged):

$$f_x(a, b) = \lim_{t \to a} \frac{f(t, b) - f(a, b)}{t - a}.$$



This is the gradient of the curve $z = f(x, b)$ at $x = a$.



Therefore, we have

- The partial derivative $f_x(a, b)$ is the slope of the tangent to the curve $z = f(x, b)$ at $x = a$.
- It gives the rate of change of $f(x, b)$ at (a, b) along the positive x-direction.
- Other notations:

$$f_x(a, b) \quad = \quad f_x(x, y)\Big|_{x=a,y=b} \quad = \quad \frac{\partial f}{\partial x}\Big|_{x=a,y=b}.$$

Example

Let $f(x,y) = x^2y^3$. Use the definition of partial derivative to compute $f_x(x,y)$.

Solution. We have

$$\begin{aligned} f_x(x,y) &= \lim_{t\to x} \frac{f(t,y) - f(x,y)}{t-x} = \lim_{t\to x} \frac{t^2y^3 - x^2y^3}{t-x} \\ &= y^3 \lim_{t\to x} \frac{t^2 - x^2}{t-x} = y^3 \lim_{t\to x}(t+x) = 2xy^3. \end{aligned}$$

We also write: $\dfrac{\partial f}{\partial x} = 2xy^3$.

Example

Consider

$$f(x,y) = \begin{cases} \dfrac{x\sin(x+y)}{x+y} & \text{if } x+y \neq 0, \\ 0 & \text{if } x+y = 0. \end{cases}$$

Determine whether $f_x(0,0)$ exists. If it does, what is its value?

Solution. We use the definition,

$$f_x(0,0) = \lim_{t\to 0} \frac{f(t,0) - f(0,0)}{t-0}.$$

Since $f(0,0) = 0$ and $f(t,0) = \dfrac{t\sin(t+0)}{t+0} = \dfrac{t\sin(t)}{t} = \sin t \quad (\text{as } t \neq 0)$, we have

$$f_x(0,0) = \lim_{t\to 0} \frac{f(t,0) - f(0,0)}{t-0} = \lim_{t\to 0} \frac{\sin t}{t} = 1.$$

• Partial derivative of $f(x, y)$ w.r.t. y

Similarly, we have partial derivative f_y.

Definition

For a function of two variables $f(x, y)$, partial derivative of f w.r.t. y is the function f_y defined as follows:

$$f_y(x, y) = \lim_{t \to y} \frac{f(x, t) - f(x, y)}{t - y},$$

(keeping x as a constant).

The subscript y indicates that the values of t tends to y.

Another notation: $\dfrac{\partial f}{\partial y}$.

The symbol $f_y(a,b)$ is the partial derivative $f_y(x,y)$ at a point (a,b) (i.e., $y=b$, keeping $x=a$ unchanged):

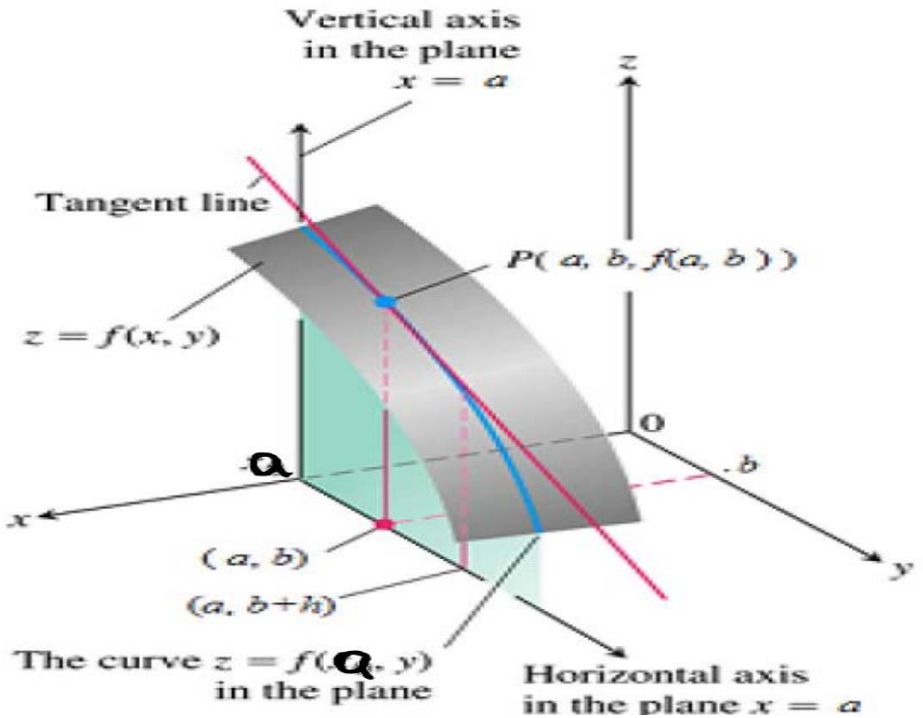
$$f_y(a,b) = \lim_{t\to b} \frac{f(a,t)-f(a,b)}{t-b}.$$

We also have

- The partial derivative $f_y(a,b)$ is the slope of the tangent to the curve $z=f(a,y)$ at $y=b$.
- It gives the rate of change of $f(a,y)$ at (a,b) along the positive y-direction.
- Other notations:

$$f_y(a,b) \quad = \quad f_y(x,y)\Big|_{x=a,y=b} \quad = \quad \frac{\partial f}{\partial y}\Big|_{x=a,y=b}.$$

This is the gradient of the curve $z = f(a, y)$ at $y = b$.



• Partial derivatives: in terms of $h \to 0$

We can also define the partial derivatives as follows:

1. Letting $t = x + h$ and noticing that $t \to x$ is equivalent to $h \to 0$, we have
$$f_x(x, y) = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}.$$

2. Similarly, with $t = y + h$, we have
$$f_y(x, y) = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}.$$

• Partial derivatives for 3-variable functions

Definition

For a function $f(x, y, z)$ of three variables, its partial derivative w.r.t. x is defined as

$$f_x(x, y, z) = \lim_{h \to 0} \frac{f(x + h, y, z) - f(x, y, z)}{h}$$

and it is obtained by regarding y and z as constants and differentiating f with respect to x.

Other partial derivatives: f_y and f_z.

• Partial derivatives of functions of n variables

More generally, we have the following definition.

Definition

For a function $f(x_1, x_2, \ldots, x_n)$ of n variables, the partial derivative w.r.t. to the variable x_i is defined by

$$f_{x_i}(x_1, x_2, \ldots, x_n) = \lim_{h \to 0} \frac{f(x_1, \ldots, x_i + h, \ldots, x_n) - f(x_1, \ldots, x_i, \ldots, x_n)}{h}.$$

Other notations: $\dfrac{\partial f}{\partial x_i}$, or simply, $f_i(x_1, x_2, \ldots, x_n)$.

• Evaluating partial derivatives

• For a fixed value of y, the definition

$$f_x(x,y) = \lim_{t \to x} \frac{f(t,y) - f(x,y)}{t - x}$$

tells us that it is the usual derivative of the function $f(x,y)$ w.r.t. x.

Therefore, to evaluate f_x, we regard y as a constant and differentiate the function $f(x,y)$ with respect to x.

Hence, we may apply rules of single-variable differentiation whenever they are applicable.

• Similarly, to evaluate f_y, we regard x as a constant and differentiate $f(x,y)$ with respect to y.

Example

Find $f_x(2,1)$ and $f_y(2,1)$, if $f(x,y) = x^3 + x^2y^3 - 2y^2$.

Solution. We have

- $f_x(x,y) = \dfrac{\partial}{\partial x}\left(x^3 + x^2y^3 - 2y^2\right) = 3x^2 + 2xy^3,$
- $f_y(x,y) = \dfrac{\partial}{\partial y}\left(x^3 + x^2y^3 - 2y^2\right) = 3x^2y^2 - 4y.$

Then

$$f_x(2,1) = 12 + 4 = 16 \quad \text{and} \quad f_y(2,1) = 12 - 4 = 8.$$

Example

If $f(x,y) = \sin\dfrac{x}{1+y}$, calculate $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$.

Solution. Using the Chain Rule for functions of one variable, we have (keep y as a constant)

$$\frac{\partial f}{\partial x} = \cos\frac{x}{1+y}\frac{\partial}{\partial x}\left(\frac{x}{1+y}\right) = \frac{1}{y+1}\cos\frac{x}{y+1},$$

and (keep x as a constant)

$$\frac{\partial f}{\partial y} = \cos\frac{x}{1+y}\frac{\partial}{\partial y}\left(\frac{x}{1+y}\right) = \frac{-x}{(y+1)^2}\cos\frac{x}{y+1}.$$

- Similarly, we have partial derivatives for 3-variable functions.

Example

For $f(x, y, z) = x \sin y + e^{3x}\sqrt{y^2 + 9z}$, compute f_x, f_y, and f_z.

Solution. We have

- $f_x(x, y, z) = \sin y + 3e^{3x}\sqrt{y^2 + 9z}$,
- $f_y(x, y, z) = x \cos y + \dfrac{e^{3x} y}{\sqrt{y^2 + 9z}}$,
- $f_z(x, y, z) = \dfrac{9e^{3x}}{2\sqrt{y^2 + 9z}}$.

Example

Find f_x, f_y, and f_z, if $f(x, y, z) = e^{xy} \ln z$.

Solution. We have

- $f_x(x, y, z) = \dfrac{\partial}{\partial x}(e^{xy} \ln z) = ye^{xy} \ln z$,
- $f_y(x, y, z) = \dfrac{\partial}{\partial y}(e^{xy} \ln z) = xe^{xy} \ln z$,
- $f_z(x, y, z) = \dfrac{\partial}{\partial z}(e^{xy} \ln z) = \dfrac{e^{xy}}{z}$.

• Second partial derivatives

• If f is a function of two variables, then its partial derivatives f_x and f_y are also functions of two variables.

Then we can consider their partial derivatives: $(f_x)_x$, $(f_x)_y$, $(f_y)_x$, and $(f_y)_y$, which are called second partial derivatives of f.

We use the notation f_{xy}, or $\dfrac{\partial^2 f}{\partial y \partial x}$ to indicate that we first differentiate f with respect to x followed by differentiating with respect to y:

$$f_{xy} = (f_x)_y \quad \text{and} \quad \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right).$$

• For a 2-variable function $f(x, y)$ there are FOUR second-order partial derivatives:

$$(f_x)_x = f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2},$$
$$(f_x)_y = f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x},$$
$$(f_y)_x = f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y},$$
$$(f_y)_y = f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2}.$$

• Similarly, for a 3-variable function, there are NINE second-order partial derivatives.

Example (Back to one of previous examples)

Find all second-order derivatives of

$$f(x, y) = x^3 + x^2y^3 - 2y^2.$$

Solution. We fist find the first-order derivatives

$$f_x = 3x^2 + 2xy^3, \quad f_y = 3x^2y^2 - 4y.$$

Now we have

$$f_{xx} = (f_x)_x = 6x + 2y^3, \quad f_{xy} = (f_x)_y = 6xy^2,$$
$$f_{yx} = (f_y)_x = 6xy^2, \quad f_{yy} = (f_y)_y = 6x^2y - 4.$$

Note that $f_{xy}(a, b) = f_{yx}(a, b)$.

• Is it true that $f_{xy} = f_{yx}$?

In general, $f_{xy} \neq f_{yx}$.

However, under some conditions, these partial derivatives are equal.

Theorem (Equality of mixed derivatives)

If $f(x, y)$ and its partial derivatives f_x, f_y, f_{xy}, and f_{yx} are defined throughout an open region containing a point (a, b) and all are continuous at (a, b), then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

The above theorem is also known as Clairaut's Theorem, named after the French mathematician Alexis Clairaut who proved it in 1740.

Example (Back to one of previous examples)

For $f(x, y, z) = e^{xy} \ln z$, find $\dfrac{\partial^2 f}{\partial z \partial y}$ and f_{zxy}.

Solution. First, we have

$$\frac{\partial^2 f}{\partial z \partial y} = \frac{\partial}{\partial z}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial z}\left(xe^{xy} \ln z\right) = \frac{xe^{xy}}{z}.$$

Next, to compute $f_{zxy} = \left((f_z)_x\right)_y$, we evaluate

$$f_z = \frac{e^{xy}}{z} \quad \Longrightarrow \quad (f_z)_x = \frac{ye^{xy}}{z}.$$

Hence,

$$f_{zxy} = \left((f_z)_x\right)_y = \frac{e^{xy} + y(xe^{xy})}{z} = \frac{e^{xy}(1 + xy)}{z}.$$

• Gradient Vector

We introduce a useful term, known as the gradient vector, denoted by ∇f. It is a vector formed by partial derivatives of f.

Definition

1. For a function f of two variables x and y, the gradient of f is the vector ∇f defined by

$$\nabla f(x, y) = (f_x, f_y) = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j}.$$

2. For a function f of three variables x, y, and z, the gradient of f is the vector ∇f defined by

$$\nabla f(x, y, z) = (f_x, f_y, f_z) = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}.$$

Example

For $f(x, y) = x(x-3) - y$, find

- $\nabla f(x, y)$
- $\nabla f(P_0)$, where $P_0 = (1, -1)$.

Solution. Since $f(x, y) = x(x-3) - y = x^2 - 3x - y$, we have

$$f_x(x, y) = 2x - 3, \quad f_y(x, y) = -1.$$

Hence,

- $\nabla f(x, y) = \big(f_x(x, y), f_y(x, y)\big) = (2x - 3, -1)$,
- $\nabla f(P_0) = \Big((2x-3)\Big|_{x=1,y=-1}, -1\Big|_{x=1,y=-1}\Big) = (2 \cdot 1, -1) = (-1, -1)$.

Example

For $F(x,y,z) = \dfrac{x^2}{4} + y^2 + \dfrac{z^2}{9}$, find the gradient vector $\nabla F(x,y,z)$.

Solution. We have

$$F_x(x,y,z) = \frac{x}{2}, \quad F_y(x,y,z) = 2y, \quad \text{and} \quad F_z(x,y,z) = \frac{2z}{9}.$$

Hence,

$$\nabla F(x,y,z) = \left(\frac{x}{2}, 2y, \frac{2z}{9}\right).$$

More generally, for a function of n variables, we have the following definition.

Definition

If f is a function of n variables $x_1, x_2, \ldots, x_n$, then the gradient vector (or simply, gradient) of f is the vector ∇f defined by

$$\nabla f(x_1, x_2, \ldots, x_n) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n} \right).$$